Raising card usage: segment-level diagnostic

Credit card activation & loyalty analysis for a Polish retail bank (fictional client)

Portfolio project, public dataset

Falling usage predicts churn: an early-warning trigger could reach 42% of churners in 23% of the base

What we found

1. Customers whose Q4 transactions fell by more than 60% vs Q1 churned at 60.0%, vs 7.1% for those with flat or rising use.

2. Attrition rises from 4.5% at 1 inactive month to 21.5% at 3 and 29.9% at 4 months.

3. Every contact with the bank adds risk: 1.8% attrition at 0 contacts, 20.1% at 3, 33.5% at 5.

The 'At risk' segment is 13.5% of customers but 37.7% of churners.

What we recommend

1. Early-warning trigger: flag customers with 3+ inactive months AND 3+ bank contacts (2,317 customers, 29.6% attrition) and the 'At risk' segment; contact them with a retention offer.

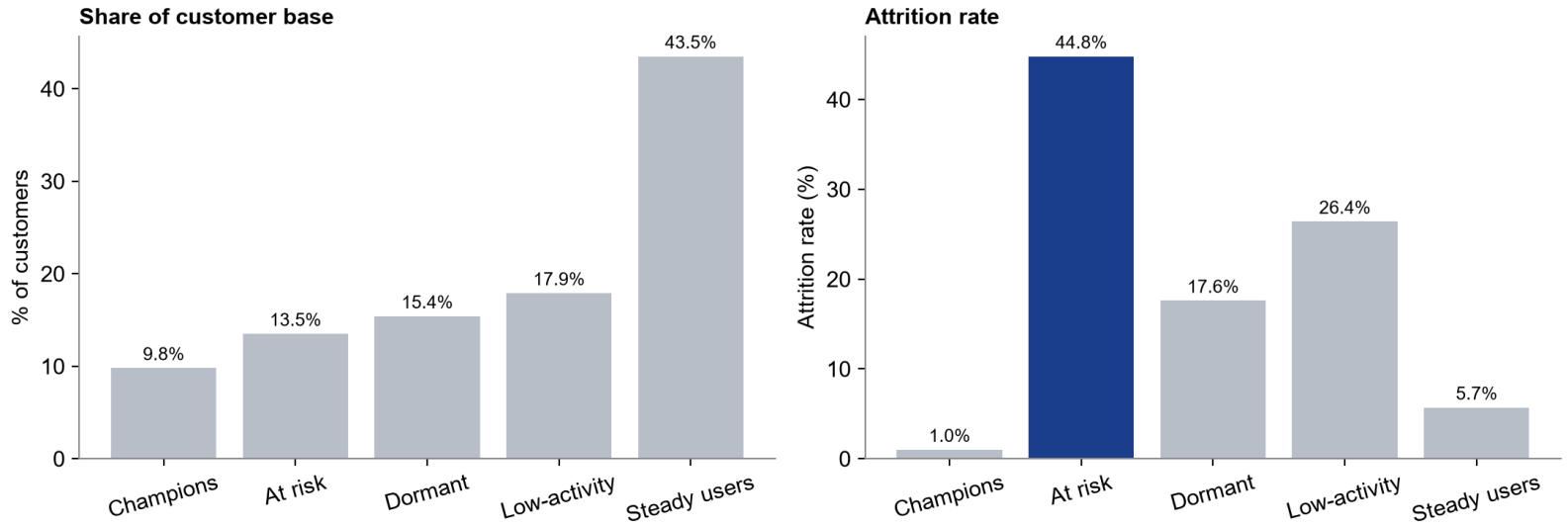
2. Activation offer for the 'Low-activity' segment (1,815 customers, 26.4% attrition): reward reaching a monthly transaction target.

3. Service-contact review: treat the 3rd contact in a year as a churn signal; find and fix the top contact reasons.

Source: Kaggle 'Credit Card Customers' (BankChurners), n = 10,127 customers; author's analysis

'At risk' customers are 13.5% of the base but 37.7% of all churners

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So what?

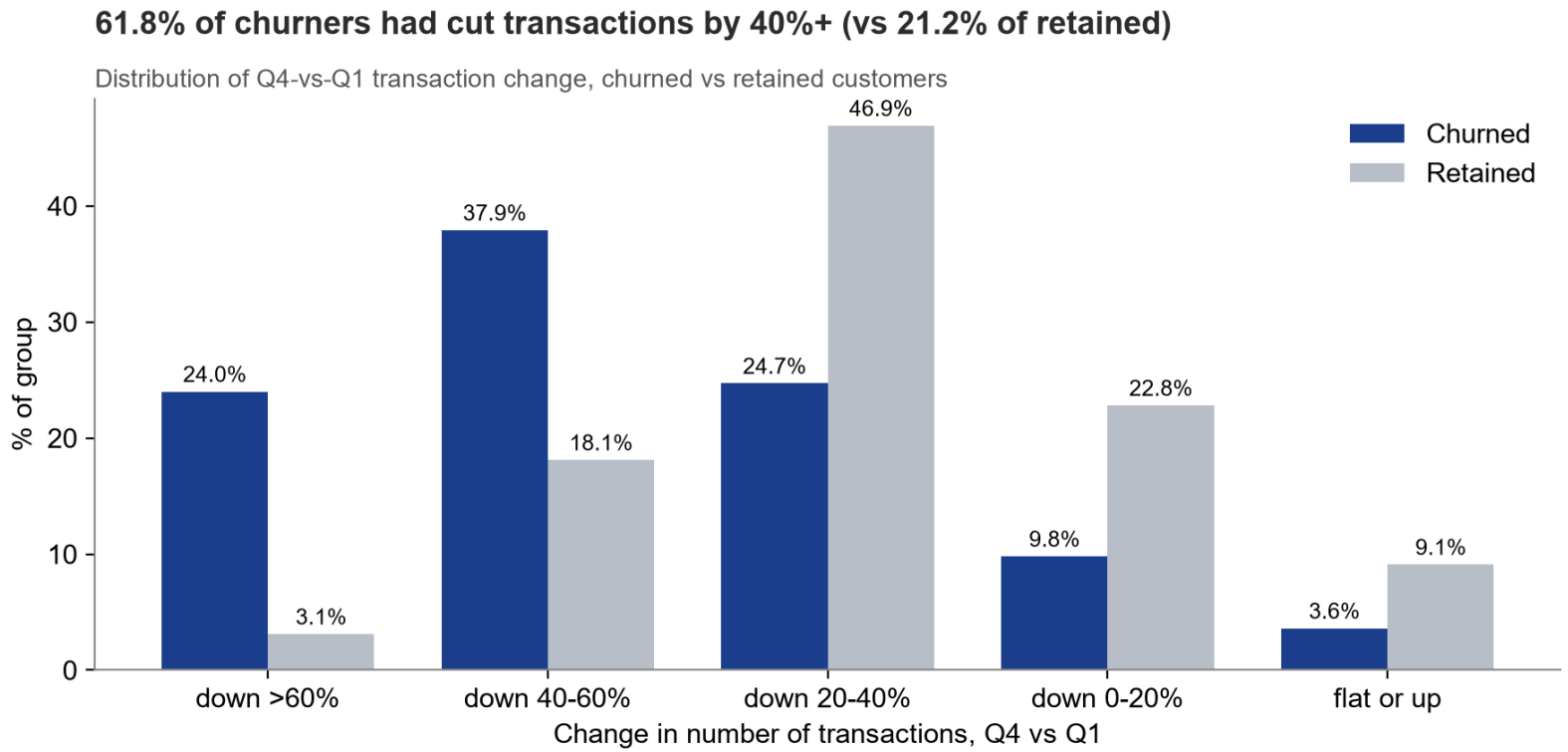
At risk: 44.8% attrition (3+ inactive months and Q4 transactions down 40%+).

Low-activity: 26.4% attrition, the second retention priority.

Champions: 1.0% attrition; Steady users: 5.7%.

Source: Kaggle 'Credit Card Customers' (BankChurners), n = 10,127; author's analysis

Finding 1: 61.8% of churners had cut transactions by 40%+ (vs 21.2% of retained)



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So what?

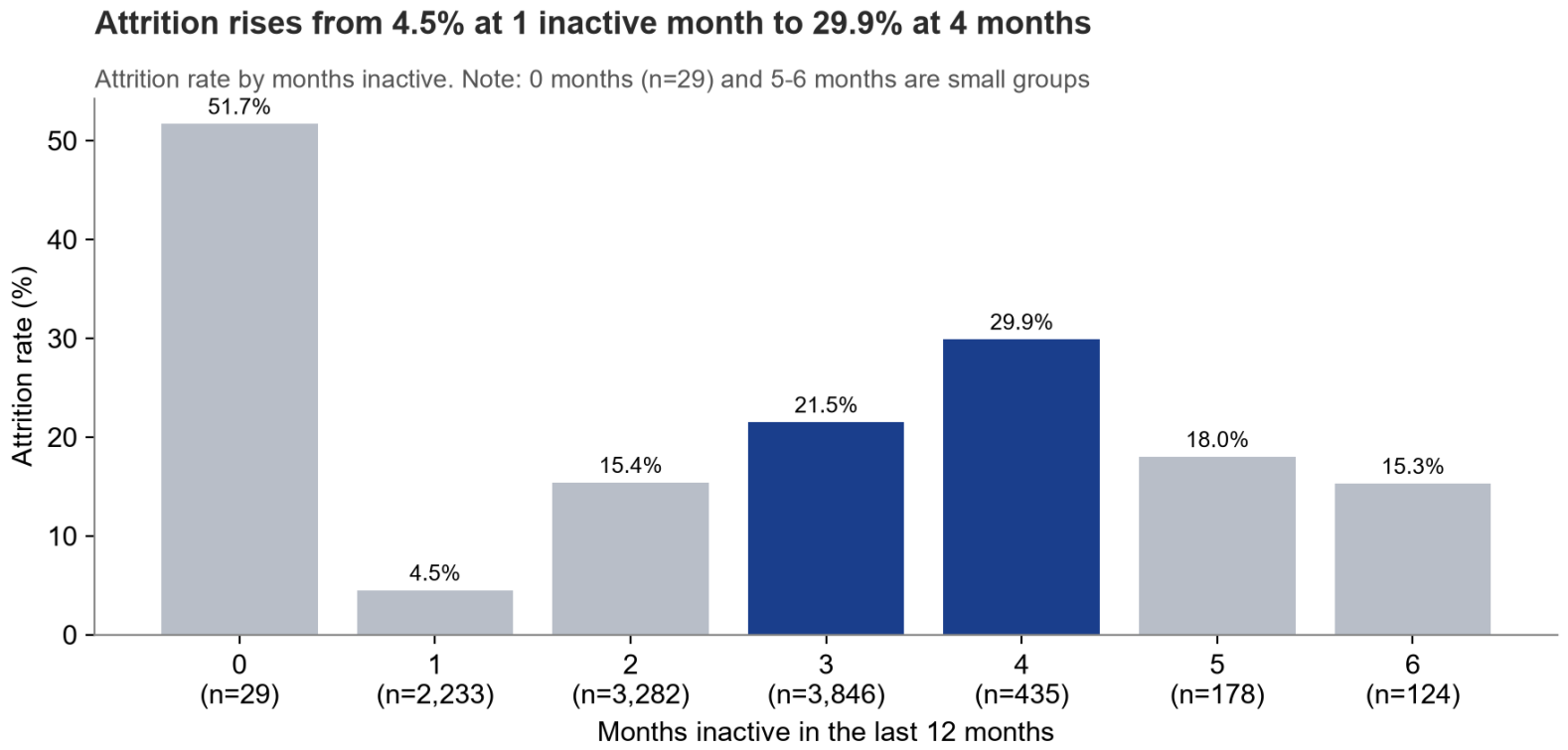
Median Q4/Q1 transaction ratio: 0.53 for churners vs 0.72 for retained.

Customers down >60% churned at 60.0% vs 7.1% when flat or up.

A drop in usage is visible in the data the bank already has: it can be monitored monthly.

Source: Kaggle 'Credit Card Customers' (BankChurners), n = 10,127 customers; author's analysis

Finding 2: attrition rises from 4.5% at 1 inactive month to 29.9% at 4 months



Source: Kaggle 'Credit Card Customers' (BankChurners), n = 10,127; author's analysis

So what?

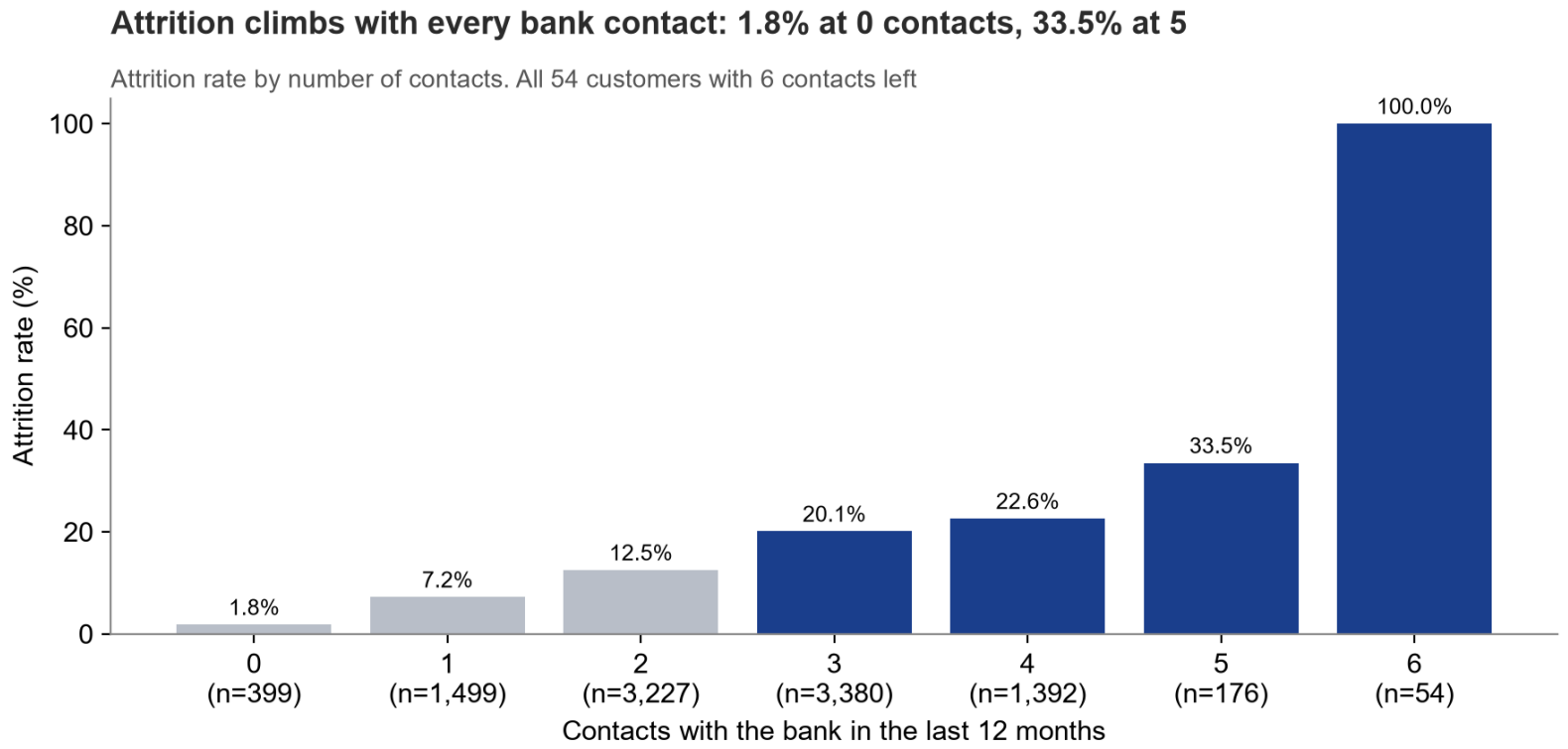
3,846 customers sit at exactly 3 inactive months (21.5% attrition): the largest pool to act on.

5-6 months and 0 months are small groups with noisy rates.

Act at month 2-3, before the customer has mentally left.

Source: Kaggle 'Credit Card Customers' (BankChurners), n = 10,127 customers; author's analysis

Finding 3: attrition climbs with each bank contact, from 1.8% to 33.5% at 5 contacts



Source: Kaggle 'Credit Card Customers' (BankChurners), n = 10,127; author's analysis

So what?

All 54 customers with 6 contacts left.

Contacts likely reflect unresolved problems (we cannot see the reason in this data).

3+ inactive months AND 3+ contacts: 22.9% of customers, 42.2% of churners, 29.6% attrition (vs 12.0% for the rest).

Source: Kaggle 'Credit Card Customers' (BankChurners), n = 10,127 customers; author's analysis

Three actions, each tested against a control group, can show impact within 6 months

Action	Target segment	Metric to track
1. Early-warning trigger + retention offer: 3+ inactive months AND 3+ contacts, or 3+ inactive months with Q4/Q1 transactions < 0.6 ('At risk')	Flagged (2,317, 29.6% attrition) and At risk (1,367, 44.8% attrition)	Attrition rate vs. control group
2. Activation offer: cashback for reaching N transactions a month for 3 months	Low-activity (1,815 customers, 26.4% attrition)	Transactions per active card; activation rate
3. Service-contact review: root-cause top contact reasons; senior handling from the 3rd contact	Customers with 3+ contacts (5,002)	Attrition rate of 3+ contact customers; repeat-contact rate

Next steps: (1) agree the trigger rule and offer budget, (2) pilot on a random half of each target group with the other half as control, (3) read results monthly for 6 months, (4) validate with the bank's own transaction-level data (exact dates, merchant categories, contact reasons).

Source: Kaggle 'Credit Card Customers' (BankChurners), n = 10,127 customers; author's analysis